AVE Sentinel: A Unified Monitoring-to-Trading Decision Support System for Early On-Chain Assets on Solana and BSC

Alex
Block Cipher
relay9a@gmail.com

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Abstract

Early-stage on-chain trading is not limited by the absence of token lists or trading terminals. The harder problem is the inability to form a trustworthy decision under severe time pressure. In the new-token, hot meme, and early-strength asset segments on Solana and BSC, traders often switch across multiple tools for discovery, contract checks, wallet tracking, liquidity inspection, and trade execution. This fragmentation creates recurring failures in evidence continuity, timing of risk checks, and replayability after a decision is made. This paper presents *AVE Sentinel*, a unified decision system that combines a primary web workspace with Telegram, CLI, and agent-facing structured runtime surfaces, organizing early-stage token work into one chain: candidate discovery, evidence verification, explainable latent scoring, risk gating, execution preparation, monitoring, and replay. AVE Sentinel is built on public AVE capabilities but reorganizes them into product modules rather than isolated endpoints. As of the market-evidence snapshot on April 10, 2026 and the implementation snapshot on April 15, 2026, the system already demonstrates the combination required by the hackathon: *Monitoring Skills*, *Trading Skills*, and a *Complete Application*. The main claim of this paper is not absolute superiority over platforms such as OKX, Binance Alpha, Bybit DEX Pro, Bitget Onchain, GMGN, or Axiom, but workflow closure through confidence-aware, risk-gated inference: AVE Sentinel turns fragmented observations into explicit action outputs such as *Proceed*, *Watch*, and *Block*. At the same time, the paper keeps a conservative implementation boundary. The current workspace shows substantial capability closure, but production-grade operating evidence still depends on wallet availability, valid credentials, approval conditions, and a larger real-case library.

The relationship with AVE is therefore complementary rather than substitutive. AVE already supplies the raw browsing layer for k-lines, holders, transactions, pairs, and risk reports. AVE Sentinel intentionally avoids becoming a heavier duplicate of those native views; its contribution is the decision layer that converts selected AVE fields into SENTINEL-8 scoring, risk gates, evidence explanations, and execution readiness.

Keywords: AVE, Solana, BSC, memecoin, on-chain monitoring, on-chain trading, risk gating, smart wallet, DEX, complete application

1 Introduction

Early-stage on-chain assets combine two properties that are often handled separately by existing tools. First, they are highly time-sensitive: price, transaction activity, wallet participation, and liquidity structure can change within minutes. Second, they are evidence-sensitive: the same surface pattern of "rapid price appreciation" can correspond to genuine traction, shallow liquidity manipulation, distribution by concentrated holders, or temporary social hype. The practical problem is not a lack of access, but a lack of a system that can turn fragmented observations into an interpretable action.

This challenge is especially visible on Solana and BSC. Both chains support active DEX ecosystems, low-friction trading, and a constant stream of new or highly speculative assets. Yet those same properties make them difficult environments for disciplined decision-making. A trader who sees a promising token on a leaderboard still needs to answer several questions before taking action: Is the liquidity thick enough to matter? Is holder concentration acceptable? Is the contract posture safe enough? Are the relevant wallets behaving like early conviction or like opportunistic short-term rotation? Existing products often provide pieces of this picture, but rarely the full chain in one place.

AVE Sentinel addresses this gap as a system design problem. Instead of building another generalized dashboard or another execution-first bot, the project organizes the user task itself and then reuses that same task logic across four entry surfaces. The target workflow is: *discover a candidate, verify evidence, perform confidence-aware risk-gated inference, preview or execute, and replay the outcome.* This design is important because the hackathon framing is not satisfied by isolated features. A credible submission must demonstrate monitoring ability, trading ability, and application completeness at the same time.

This also defines the boundary of the product. AVE Sentinel is not an attempt to replace AVE's own token pages, chart views, holder pages, or transaction browsers. Those surfaces remain better suited for detailed raw inspection. The system keeps only the fields required for judgement and dedicates more product space to the model: the eight SENTINEL-8 factors, weighted contribution, risk suppression, and the final decision label.

2 Background and Problem Definition

2.1 The problem is not missing data, but missing decision structure

The typical early-stage on-chain workflow is fragmented across several interfaces. A user starts from a trending page or a "new pairs" page, jumps to another tool for price or liquidity inspection, opens a wallet tracker to see whether smart money has entered, and finally returns to a trading terminal for a quote or a fast buy. This fragmentation creates three structural problems.

1. **Evidence discontinuity.** Data is visible, but the relationship among signals is left to user intuition. Existing interfaces rarely explain why a set of observations should justify *Proceed*, *Watch*, or *Avoid*.

2. **Risk checks happen too late.** In many execution-centric tools, speed comes before structure. Users are pushed closer to a buy button before they have completed holder, contract, or liquidity checks.

3. **Outcomes are hard to replay.** The evidence visible at the moment of a decision is often not preserved in a reusable, comparable form, which weakens post-trade learning.

2.2 Why ordinary token discovery tools and ordinary bots are not enough

Trending boards answer "what is hot right now" but usually do not answer "why this is actionable" or "why this should be blocked." Ordinary token discovery interfaces can accelerate visibility, but they do not necessarily connect smart wallet evidence, holder structure, contract risk, and execution readiness. Ordinary trading bots reduce latency, but often move users closer to making a fast mistake rather than a grounded one. For the new-token and meme-token setting, the relevant objective is therefore not faster clicking in isolation; it is more reliable decision support under fast-moving conditions.

3 Market and Security Drivers

3.1 Why Solana and BSC are a justified entry point

As checked on April 10, 2026, DefiLlama reports approximately $2.882B in 24-hour DEX volume and $71.967B in 30-day DEX volume on Solana, while BSC reports approximately $824.94M in 24-hour DEX volume and $29.768B in 30-day DEX volume [17, 18]. These values indicate that both chains remain dense enough in terms of opportunity generation and tradeability. For an early-stage asset system, this is a practical sweet spot: the domain is large enough to matter, but still narrow enough to support a complete product story rather than an unfocused all-chain tool.

3.2 Why new tokens, hot memes, and early-strength assets matter

As checked on April 10, 2026, CoinGecko reports roughly $34.3B in market capitalization and $3.19B in 24-hour volume for the meme token category, while the Four.meme ecosystem on BNB Chain stands at roughly $748M in market capitalization [19, 20]. These figures do not prove quality, but they show that meme and early-stage assets are not a negligible fringe. They are a meaningful, liquid, and attention-heavy submarket. The issue is that their information half-life is short, which makes a structure-first system more valuable than a generic watchlist.

3.3 Why risk gating must be central

CertiK estimates roughly $3.352B in Web3 losses for 2025, including roughly $1.451B associated with supply chain attacks [21]. Chainalysis estimates roughly $17B in crypto scam and fraud losses in 2025 and reports approximately 1400% year-over-year growth in impersonation scams [22]. These figures do not map one-to-one onto memecoin trading losses, but they are sufficient to support a design principle: security and structural risk are not optional overlays. They are first-class constraints in any system that claims to support real on-chain action.

4 System Goals and Design Principles

AVE Sentinel is not trying to become a larger platform than every competitor. Its objective is more specific: to become a more complete decision chain for a narrow, high-noise, high-speed scenario. Five principles follow from that objective:

1. **Evidence should outrank heat.** Leaderboards are inputs, not conclusions.
2. **Risk gating should precede execution.** A failed structural check should stop the workflow before quoting or swapping.

3. **Smart money is supporting evidence, not a substitute for judgment.** Wallet participation can strengthen confidence but cannot neutralize contract or liquidity risk.

4. **Trading must return to monitoring.** Execution is not the end of the workflow; the system must move into position watching and replay.

5. **Implemented, scaffolded, and pending capabilities must be separated.** A hackathon paper should not present pending integrations as finished production features.

4.1 Decision abstraction

For a candidate asset x, AVE Sentinel does not map price movement directly into an action. It first builds five forms of evidence:

$$\mathcal{E}(x) = \{M(x), S(x), R(x), W(x), X(x)\},$$

where $M(x)$ denotes market activity and momentum evidence, $S(x)$ denotes structural evidence such as liquidity and holder concentration, $R(x)$ denotes contract and risk evidence, $W(x)$ denotes wallet-quality evidence, and $X(x)$ denotes execution evidence such as quote feasibility and expected slippage. The system then emits a discrete decision

$$D(x) \in \{\texttt{Proceed}, \texttt{Watch}, \texttt{Block}\}.$$

In this design, $R(x)$ and $S(x)$ act as hard gates, while $M(x)$ and $W(x)$ contribute confidence. Execution evidence $X(x)$ is only considered after the risk and structure gates have been cleared.

5 AVE Sentinel System Architecture

From the product perspective, AVE Sentinel is a four-surface system. The web interface is the primary analysis workspace; Telegram provides a command-first operational surface; CLI provides a scriptable operational surface; and the agent-facing skill runtime exposes the same decision logic through a stable structured JSON contract. The main end-to-end workflow remains:

`Overview / Radar` → `Score Model` → `Token Dossier` → `Risk Guard` → `Opportunity Desk` → `Replay / Docs`

Internally, the current workspace exposes eight visible surfaces:

- **Overview:** product thesis, workflow strip, and multi-entry orientation.
- **Docs:** in-product product guide, command reference, and demo narrative.
- **Access:** user-tier and rollout logic for future gated usage.
- **Radar:** candidate staging board and verdict surface.
- **Token Dossier:** one-page evidence view for token and pair structure.
- **Score Model:** SENTINEL-8 signals, latent score, confidence, and verdict output.
- **Risk Guard:** contract posture, concentration, and liquidity-event interpretation.

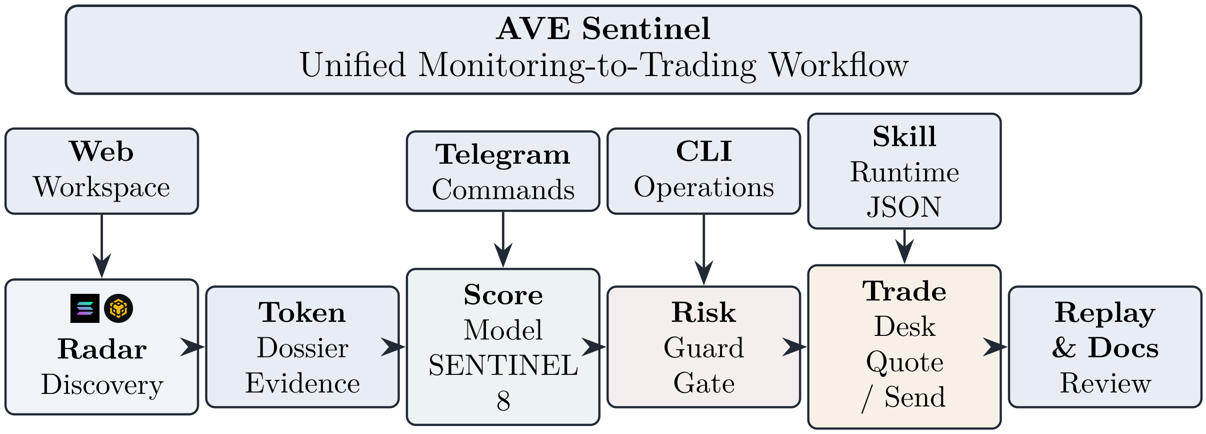

Figure 1: AVE Sentinel combines four entry surfaces while keeping one shared decision workflow. The web workspace is the deepest analysis path; Telegram, CLI, and the skill runtime reuse the same product core.

- **Opportunity Desk:** manual chain-wallet and delegate-wallet execution paths.

Replay remains present, but it is currently carried as an embedded event timeline and decision-outcome trace tied to token detail and trade actions rather than as a standalone top-level page.

This architecture mirrors the current workspace. The current repository shows one shared product core across Web, Telegram, CLI, and the skill runtime, while the proxy layer routes not only data and chain-wallet trade calls but also delegate-wallet requests that require AVE signatures.

The skill runtime is important because it is not merely an internal helper script. The current workspace exposes it as a stable JSON runtime with one shared response envelope, bilingual human-readable messages, and explicit skill groups for `data`, `chain_wallet`, and `delegate_wallet`. This means the same product logic can be consumed by an agent without screen scraping or prompt-only routing, which materially strengthens the project's claim to be a complete application rather than only a browser demo.

6 Core Module Design

6.1 Overview and Docs

Overview is not merely decorative. In the current product it states the system thesis, summarizes the workflow, and makes the multi-entry nature of the system explicit. Docs plays a complementary role: it embeds command surfaces, environment assumptions, and a recommended demo path directly inside the product. Access adds a third global layer by making explicit how future rollout, quota, and tiered privileges could be attached to the same research-and-trading stack. For a hackathon submission this matters because completeness is partly communicative; a product that explains its own workflow and access boundary is easier to evaluate and defend than one that leaves operational context scattered across repository files.

6.2 Radar

Radar is not intended to be a bigger leaderboard. It is a candidate staging area. In the current implementation, the module supports chain filtering, verdict filtering, and keyword search over candidate rows. When an AVE API key and the local proxy are available, Radar is populated from live trending data; when they are not, the workflow remains demonstrable through mock or fallback payloads. The live surface currently draws from Solana, BSC, Base, and Ethereum, while the paper's main evaluation narrative remains centered on Solana and BSC. Candidate ordering and the `Proceed / Watch / Block` labels are not produced by ad-hoc heuristics but by the SENTINEL-8 scoring model described in Section 7, which is the main algorithmic contribution of this paper.

6.3 Token Dossier

The dossier compresses token-level evidence into one page: token identity, price change, volume, market cap or FDV, main pair structure, liquidity, risk status, holder structure, and signal-linked pair context. Its purpose is not to reproduce every AVE-native chart, holder row, or transaction row inside Sentinel. It selects the evidence needed to support the score model and action gate, while users can still return to AVE for full raw inspection. The dossier exists because users need a stable, replayable object for answering a practical question: "Is this worth carrying forward to deeper review?"

6.4 Score Model

The scoring surface is only useful when it explains how a verdict is produced. AVE Sentinel therefore exposes SENTINEL-8 as a first-class product module rather than hiding it behind a single score badge. The current implementation shows interpretable evidence signals, a compact latent score, a confidence-aware verdict path, and explicit risk suppression on a dedicated page. This turns the score from a cosmetic rank into an inspectable decision object that can be checked before a user moves into Token Dossier, Risk Guard, or the Opportunity Desk.

6.5 Risk Guard

Risk Guard is the central module that turns analysis into action constraints. Its design target is not just to present a score, but to emit an action-level recommendation. In the current live implementation, liquidity transactions are already pulled into the risk layer and used to distinguish between recent additions and removals of liquidity, while holder concentration is currently derived from the contract-risk payload rather than from a separate dedicated holders endpoint. This distinction matters because it shows progress without overstating endpoint completeness.

6.6 Opportunity Desk

The trading layer is designed around explainable action rather than automation for its own sake. In the current workspace the Opportunity Desk exposes two concrete paths. The first is a manual chain-wallet path: official AVE quote, official auto-slippage and gas-tip hints, official Solana or EVM transaction build, local-wallet signing, signed-transaction submission back to AVE send endpoints, and status refresh where supported. The second is a delegate-wallet path: delegate-wallet listing or creation, approval lookup, market or limit order submission, and order-status refresh. At the frontend level, the workspace also exposes a Privy-based multi-chain wallet-connect surface with chain-aware wallet options, plus switch and disconnect handling, while keeping execution user-controlled rather than custodial. This is a stronger fit for the hackathon than an execution-only narrative because it shows how trading ability emerges from monitoring evidence rather than bypassing it.

6.7 Structured Skill Runtime

The new skill layer turns the project into an agent-callable system rather than a UI-only workspace. The runtime can list skills, publish JSON schemas, and execute named operations across three grouped areas: data lookup (`radar_scan`, `token_dossier`, `risk_guard`); chain-wallet trading (`trade_quote`, `trade_build_unsigned`, `trade_send_signed`); and delegate-wallet execution (`delegate_wallet_list`, `delegate_wallet_create`, `delegate_market_order`, `delegate_limit_order`). Because these calls all return the same structured envelope, the runtime acts as a formal reuse layer for the same evidence and trading chain already present in the web product.

6.8 Replay, Docs, CLI, Telegram, and Skill Surfaces

A complete application must preserve why a decision was made and what happened afterward. Replay now records not only static prototype notes but also action-side events written back from the Opportunity Desk, including approval, build, manual-send, and delegate-order records. Docs, CLI, Telegram, and the structured skill runtime add a second layer of completeness: the system is not web-only, but exposes the same candidate, risk, quote, wallet, approval, and order logic across a product guide and three operational surfaces backed by the same shared core. The command surfaces now extend beyond read-only queries and include practical actions such as `/wallets`, `/wallet`, `/approve`, and `/order`, while the skill runtime exposes the same state transitions through structured calls rather than natural-language prompting. Replay still needs more real-case depth, but the current structure already makes the end-to-end application story stronger than a passive analysis page.

7 SENTINEL-8: An Eight-Signal Risk-Gated Inference Model

Early-stage candidate selection is not a pure ranking problem. In the new-token and hot-meme setting, the system must answer three questions simultaneously: whether an asset is active enough to matter, structurally credible enough to review further, and risky enough to suppress despite short-term attention. SENTINEL-8 is introduced as an eight-signal latent scoring model that keeps the observable evidence legible while moving the actual inference step into a more compact risk-aware representation.

7.1 Model Intuition

Let x denote a token candidate. SENTINEL-8 starts from eight interpretable signals,

$$\Phi(x) = (L, V, M, A, C, R, S, F),$$

where L denotes liquidity depth, V volume quality, M momentum, A activity acceleration, C holder concentration, R risk posture, S smart-money signal quality, and F freshness. The role of these signals is explanatory as much as predictive: they preserve a human-readable bridge from raw AVE fields to the eventual verdict shown on Radar and the Score Model page.

Rather than treating those signals as eight unrelated knobs, the model groups them into four evidence views: market state, structural quality, explicit risk, and wallet-context evidence. In the notation of Section 4, this corresponds to a compact mapping from the abstract evidence set $\mathcal{E}(x)$ into a score that remains interpretable but no longer reads like an implementation checklist.

Signal	Evidence view	Interpretive role
L	Structure	Tradable depth and pool thickness under real execution constraints.
V	Market	Whether notional flow is substantial enough to support attention.
M	Market	Short-horizon and daily directional strength without reducing the model to heat alone.
A	Market	Whether transaction activity is accelerating rather than merely accumulated.
C	Structure	Ownership dispersion and concentration pressure.
R	Risk	Contract and safety posture that can suppress otherwise attractive candidates.
S	Wallet / Context	Quality of signal-linked wallet participation and informed-flow evidence.
F	Wallet / Context	Temporal positioning of the asset inside the early-stage opportunity window.

Table 1: SENTINEL-8 interpretability map. The eight signals remain visible to the user, while the inference layer operates on a smaller set of latent evidence views.

7.2 Latent Representation

Let $\mathcal{V}_x = \{v_M(x), v_S(x), v_R(x), v_W(x)\}$ denote the four evidence views for candidate x. Each view is encoded by a learned or calibrated embedding map

$$e_j(x) = f_j(v_j(x)), \qquad j \in \{M, S, R, W\},$$

and the candidate is represented in a shared latent space by

$$z_x = [e_M(x); e_S(x); e_R(x); e_W(x)] \in \mathbb{R}^d.$$

This formulation keeps the score model compact. The observable signals still provide interpretability, but the actual inference step is performed on a fused representation that can absorb cross-view interactions such as "strong momentum, but weak structure" or "acceptable structure, but prohibitive risk."

7.3 Risk-Gated Scoring

Risk is not treated as just another additive feature. Instead, the risk view modulates the usable evidence budget before the final score is formed. Let r_x denote the latent risk summary derived from $e_R(x)$. The gated representation is

$$G(r_x) = \sigma(W_g r_x + b_g), \qquad h_x = G(r_x) \odot z_x,$$

where $\odot$ denotes element-wise gating. The scalar decision score is then

$$s_x = w^\top h_x + b,$$

and the corresponding action probability is

$$p_x = \sigma\big(\alpha(s_x - \tau)\big).$$

The model emits a three-way decision surface rather than a single leaderboard rank:

$$D(x) = \begin{cases} \texttt{Proceed}, & p_x \geq \tau_p \text{ and } \kappa_x = 1, \\ \texttt{Block}, & p_x < \tau_b \text{ or } \kappa_x = 0, \\ \texttt{Watch}, & \text{otherwise,} \end{cases}$$

where κ_x denotes hard-stop eligibility from contract and liquidity checks. In product terms, this is why SENTINEL-8 is better read as structured inference under uncertainty than as a cosmetic score. Strong momentum can still be suppressed by structural or contract evidence, while moderate momentum can become actionable if the rest of the evidence stack is coherent.

7.4 Training / Calibration Objective

SENTINEL-8 is designed to be calibratable from replay cases without requiring the paper to claim a fully online-trained production model. Let y_x denote action validity and let ordered pairs (i, j) encode that candidate i should outrank candidate j under replay evidence. The compact objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \mathcal{L}_{\text{rank}} + \gamma \mathcal{L}_{\text{cal}},$$

where $\mathcal{L}_{\text{cls}}$ aligns p_x with action-validity labels, $\mathcal{L}_{\text{rank}}$ preserves relative ordering among replay cases, and $\mathcal{L}_{\text{cal}}$ keeps reported confidence numerically consistent with observed outcomes. A standard ranking term is

$$\mathcal{L}_{\text{rank}}(i, j) = \log\big(1 + \exp\big(-(s_i - s_j)\big)\big).$$

This objective is intentionally compact. It says that the model should both classify and rank, while remaining confidence-aware enough to support real action thresholds. The current prototype should therefore be read as a calibratable inference design with interpretable signals and a stable decision contract, not as an already fully trained black-box selector.

In deployment, chain-local threshold balancing can still be applied at the surface layer so that `Proceed`, `Watch`, and `Block` remain informative under different chain distributions. In this paper, that balancing is treated as a presentation or deployment policy rather than as the mathematical core of the model itself. The same decision tuple can then be reused across Web, Telegram, CLI, and the structured skill runtime, which is precisely why SENTINEL-8 contributes to Monitoring Skills, Trading Skills, and Complete Application at the same time.

8 Mapping to AVE Official Capabilities

8.1 Publicly verifiable AVE capabilities

As of April 10, 2026, the official AVE introduction page states coverage of more than 160 blockchains, more than 300 DEXs, and more than 7 million users [1]. The public API v2 documentation directly exposes token search, price reads, token detail, pair or token k-lines, top holders, pair transactions, trending lists, and contract risk reports [2]. These capabilities are already enough to support the first level of AVE Sentinel: candidate discovery, token and pair inspection, holder concentration review, and contract risk gating.

Because those capabilities already exist in AVE, AVE Sentinel does not treat raw k-lines, full holder tables, or transaction feeds as its main product contribution. The project uses them as inputs and evidence anchors, then focuses the user interface on model interpretation, score formation, risk gating, and execution preparation.

8.2 Product-side monitoring and trading signals

Beyond API documentation, AVE's own educational and product-facing materials support the broader monitoring-to-trading thesis. The safety tutorial shows that AVE surfaces warnings for unaudited or high-risk tokens [3]. The Signal Center blog frames smart wallet activity, abnormal volume, market cap acceleration, and social momentum as actionable discovery signals [4]. The AVE Sniper Bot documentation on Solana shows a product-side trading flow with rapid buy or sell, anti-MEV mode, and post-trade PnL reporting [5]. The point is not that every capability is exposed in identical public API form, but that AVE already spans both monitoring-side and trading-side product logic.

8.3 Why AVE is a suitable base layer

AVE is suitable for AVE Sentinel not because it guarantees superiority on every dimension, but because it provides a compatible substrate for a complete early-stage asset workflow. Its official materials cover market reads, risk inspection, wallet and signal interpretation, and product-side execution paths. The current project structure now reflects that suitability more directly by using both AVE data endpoints and AVE trade-oriented routes through a local proxy and an execution-preparation layer. This is materially different from building a workflow on top of tools that only solve discovery, only solve execution, or only solve wallet tracking.

9 From Monitoring to Trading: The Complete Workflow

The expression *Complete Application* is meaningful only if a user can complete a full task inside the product. For AVE Sentinel, that task is:

1. **Discover** candidates from Solana and BSC trending or early-stage pools.
2. **Verify** token, pair, wallet, holder, liquidity, and contract evidence.
3. **Decide** among `Proceed`, `Watch`, and `Block`.
4. **Preview or prepare execution** using official quote flows, slippage or gas hints, transaction build, wallet signing, signed send, or delegate-order submission.

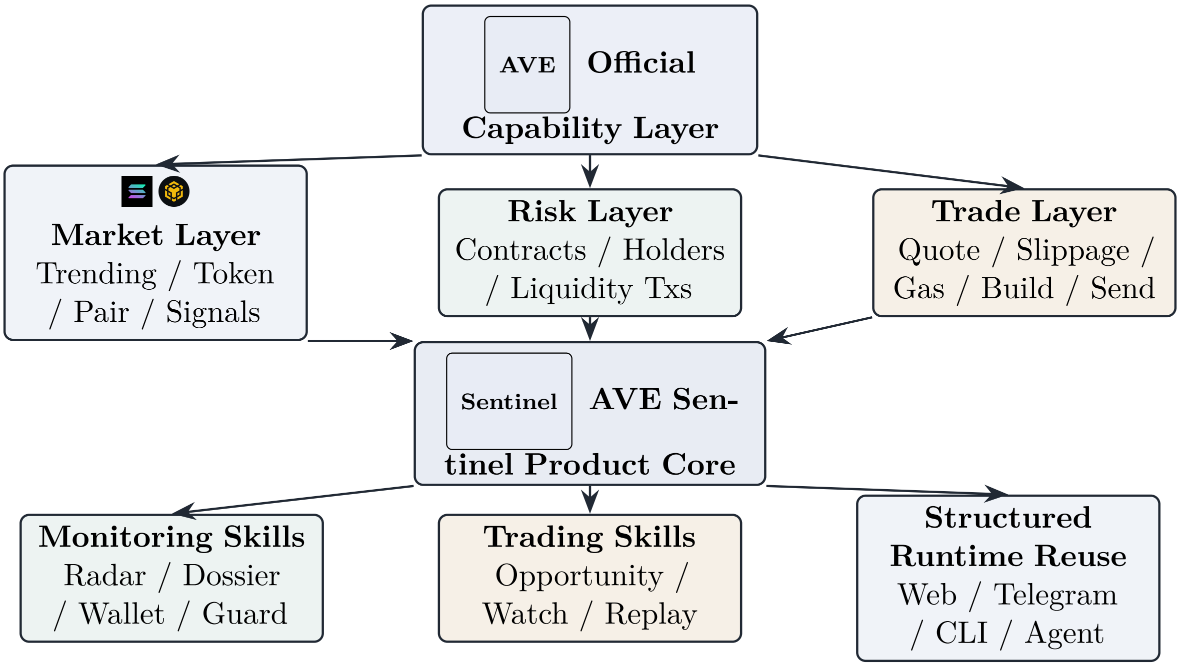

Figure 2: Capability mapping from AVE to AVE Sentinel. The project uses AVE as a base layer, then reorganizes it into monitoring modules, trading modules, and a reusable structured runtime layer.

5. **Monitor** post-entry risk and wallet behavior.
6. **Replay** the original evidence and compare it to the outcome.
7. **Reuse the same decision contract** through Telegram, CLI, or the structured skill runtime.

This is also why the project must be described as satisfying three requirements simultaneously:

- **Monitoring Skills:** risk, wallet, and structure interpretation.
- **Trading Skills:** signal generation, official Dry Run, wallet-controlled signing or delegate-order submission, position watching, and replay.
- **Complete Application:** a continuous workflow plus reusable structured interfaces rather than isolated screens.

The current implementation makes this claim stronger than before because the same workflow is no longer confined to one web page. The web workspace remains the deepest analytical surface, but CLI, Telegram, and the skill runtime already expose Radar, token, risk, quote, wallet, approval, and order actions through the same shared core, so the product behaves like one system with multiple operational surfaces rather than as unrelated demos.

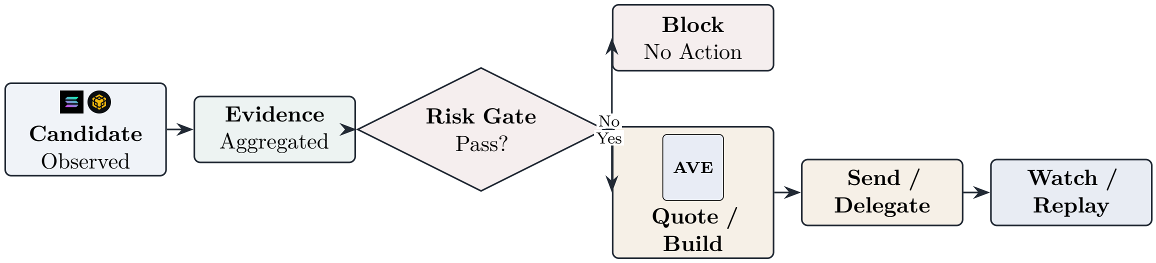

Figure 3: Monitoring-to-trading decision loop. Execution is entered only after the risk gate is cleared, then returns to watch and replay.

10 Comparative Discussion: AVE Sentinel versus Single-Point Tools

10.1 Comparison boundary

This paper does not claim that AVE Sentinel outperforms every competing platform across all contexts. The relevant question is narrower: what portion of the decision chain does each product primarily solve?

10.2 Platform boundary differences

- **AVE** provides the base layer: multi-chain data, risk inspection, and product-side trading flows [1, 2, 5]. It is the infrastructure and capability substrate rather than a narrow Solana+BSC decision product by itself.
- **OKX DEX** is a broad discovery-plus-execution aggregator. The official guide emphasizes multi-chain aggregation, cross-chain trading, Explore discovery, token profiles, liquidity metrics, and chain-level indicators [6]. This makes it a strong general entry point, but not a dedicated evidence-closure system for a narrow early-stage scenario.
- **Binance Alpha** is oriented toward early project exposure inside Binance Wallet and lowers purchase friction through Quick Buy, automated slippage handling, and anti-MEV protection [7]. This is valuable for access, but it does not by itself constitute a replay-centered decision workflow.
- **Bybit DEX Pro** emphasizes data-enhanced trading. Its i-SMART stack includes influential trader tracking, social checks, real-time trending indicators, and token safety assessments [9]. This validates the market demand for analysis-enhanced DEX terminals, yet it still differs from a complete monitoring-to-replay product.
- **Bitget Onchain** reduces the barrier for CEX users to access on-chain assets directly from a spot-account context [8]. The main contribution is operational convenience, not evidence organization.
- **GMGN** is highly optimized for meme-token discovery, smart money tracking, wallet radar, copy trading, and fast action, but its own documentation still tells users to check contract safety carefully before following trades [10, 11, 12]. This makes GMGN an excellent speed-oriented memecoin terminal, but not a complete risk-gated decision system.

- **Axiom** is a high-intensity Solana trading environment with strong token discovery, trader-level wallet scans, and structural filters such as liquidity and top-holder percentages [13, 14, 15, 16]. It is closer to an execution-centric Solana terminal than to a dual-chain monitoring-to-trading application.

10.3 Central comparative claim

The distinct contribution of AVE Sentinel is therefore not a longer feature list. It is the organization of fragmented tasks into a single workflow with evidence pages and explicit action gates. In the new-token and hot-meme setting, that integration matters more than another leaderboard or a marginally faster buy path.

11 Related Platforms and Competition

Existing platforms can be grouped into four broad categories: general aggregation and wallet execution platforms such as OKX DEX and Bitget Onchain; early-stage project exposure products such as Binance Alpha; meme and smart-money terminals such as GMGN and Axiom; and data-enhanced DEX terminals such as Bybit DEX Pro. Each category solves part of the problem. What is still missing is a system that lets a user see heat, structure, risk, and action guidance at the same time, then preserve that evidence for replay.

12 Case Studies

The following two cases are system walkthroughs derived from the current prototype and mock data in the workspace. They illustrate mechanism rather than validated live trading performance.

12.1 Case A: Opportunity discovery without blind chasing

The prototype candidate `MOONX` on Solana is treated as a positive candidate for a small-size dry run. The reasons are not merely directional price movement. The mock evidence stack shows synchronized activity in price and transaction flow, approximately $820K in main-pair TVL, an acceptable risk posture, approximately 21.4% concentration among the top ten holders, no obvious continuous liquidity removal in the recent window, and participation from a small number of higher-quality wallets. In the current product, that path now extends beyond static analysis: the same candidate can move into official quote, hint retrieval, transaction build, and replay logging inside one surface. This does not prove inevitable upside. It shows why the system can rationally move the asset into a `Proceed with size control` state instead of either ignoring it or labeling it a guaranteed win.

12.2 Case B: Risk interception before execution

The prototype candidate `FROGZ` on Solana appears hot at the surface level, with high attention and sizable 24-hour activity, yet the structural evidence is poor. The mock evidence shows very shallow liquidity relative to the attention profile, top-ten holder concentration near 58.7%, recent liquidity removal, and a pattern more consistent with a sharp hype pulse than with stable continuation. In this case the correct output is not `Watch` but `Block`. The point of the case is to show that a serious system must sometimes stop the user from acting, even when the market surface still looks attractive.

13 Limitations and Future Work

The current workspace snapshot must be described with precision.

1. The workspace already includes a shared core across Web, Telegram, CLI, and the structured skill runtime; a proxy for AVE data, trade, and delegate routes; live Radar enrichment; official quote and hint retrieval; manual chain-wallet signing and signed-send paths; delegate-wallet approval and order paths; a Privy-based external wallet-connect surface for both EVM and Solana; and replay records written from trade actions.

2. These paths are implemented at code and UI level, but this paper still does not claim validated live performance. Successful real use depends on wallet availability, valid AVE credentials, token-specific approval conditions, and market conditions that the current prototype has not benchmarked systematically. In particular, some EVM sell-side situations can still require an `approve-chain` step outside the shortest demo path.

3. The current case studies are prototype walkthroughs rather than statistically supported live-trading claims.

4. The current product no longer exposes a dedicated wallet-intelligence page. Signal-linked wallet context still contributes to SENTINEL-8 and token detail where available, but public wallet coverage is not yet stable enough to justify a standalone surface.

5. The paper's main narrative stays on Solana and BSC even though the current runtime surface already extends Radar and trade tooling to Base and Ethereum.

6. Replay is currently event-log oriented rather than a mature case library with large-sample outcome analysis.

7. The skill runtime is already stable enough to expose schemas and structured calls, but it still inherits the same backend readiness constraints as the UI surfaces and should not be mistaken for a fully benchmarked production API.

The next practical steps are therefore straightforward: deepen live validation of the existing execution paths, harden approval handling across more EVM token categories, expand Replay into a structured case library, refine token-specific wallet joins further, strengthen the skill-runtime contract around more live case paths, and narrow the gap between the paper's Solana+BSC focus and the current wider runtime surface.

14 Conclusion

AVE Sentinel is best understood as an evidence-grounded decision support system for early on-chain assets on Solana and BSC. It is not merely another data dashboard, and it is not merely a fast-trading bot. Its contribution is to organize candidate discovery, explainable latent scoring, token evidence review, risk interception, quote or execution preparation, monitoring, and replay into one coherent application, then expose that same logic consistently across Web, Telegram, CLI, and an agent-facing structured runtime. This is what allows the project to satisfy Monitoring Skills, Trading Skills, and Complete Application at the same time.

The paper deliberately avoids unsupported claims about absolute platform superiority or guaranteed profit improvement. The more defensible conclusion is narrower and stronger: in early-stage token trading, the missing product is often not another feed, but a complete workflow that turns fragmented observations into structured inference under uncertainty and then into explicit, replayable decisions. AVE Sentinel is a direct response to that gap.

A Evidence Table

ID	Claim	Evidence	Source and Date	Used In	Status
E01	AVE Sentinel is designed as a full decision loop rather than a generic dashboard.	Repository documents and in-product Docs define the workflow as discovery, evidence review, judgment, preview or execution, monitoring, and replay.	Repository inspection, checked Apr. 15, 2026.	Intro; Architecture; Conclusion	Strong
E02	The project is intentionally framed as Monitoring Skills, Trading Skills, and Complete Application at the same time.	The repository narrative maps score and risk explanation to monitoring, Opportunity Desk to trading, and the full workflow plus reusable runtime to application completeness.	Repository inspection, checked Apr. 15, 2026.	Intro; Complete Workflow	Strong
E03	The current workspace already contains a shared core across Web, Telegram, CLI, and the structured skill runtime, plus a local AVE proxy.	README documents the multi-surface architecture, while the proxy script routes data, trade, and delegate requests with AVE signature handling, and the skill runtime exposes the same core through named structured calls.	Repository inspection, checked Apr. 15, 2026.	Architecture; Limitations	Strong
E04	Official quote, manual signing paths, signed-send, and delegate-order flows are already implemented, but live validation depth is still limited.	The trade and wallet layers implement official quote, auto-slippage, gas-tip, transaction build, local-wallet signing, signed-send calls, delegate approval, market or limit order submission, and order-status refresh.	Repository inspection, checked Apr. 15, 2026.	Architecture; Opportunity Desk; Limitations	Strong
E05	The product already exposes the major workflow modules in the UI.	The React app contains Overview, Docs, Access, Radar, Score Model, Token Dossier, Risk Guard, and Opportunity Desk surfaces, while replay remains available as an embedded timeline rather than as a standalone top-level tab.	Repository inspection, checked Apr. 15, 2026.	Core Modules	Strong
E06	The current data layer already targets AVE live endpoints with fallback logic, live enrichment, and deterministic ranking.	The integration layer requests live trending, token detail, contract risk, signal-linked wallet context, liquidity transactions, and SENTINEL-8 factor inputs across Solana, BSC, Base, and Ethereum, while the trade layer handles quote, hints, build, send, and delegate flows with fallback behavior.	Repository inspection, checked Apr. 15, 2026.	Architecture; Core Modules; SENTINEL-8	Strong
E07	AVE has sufficient scale to serve as a platform substrate.	The official introduction page states coverage of 160+ blockchains, 300+ DEXs, and 7M+ users.	AVE introduction page, checked April 10, 2026.	AVE Capability Mapping	Strong

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ID	Claim	Evidence	Source and Date	Used In	Status
E08	AVE publicly exposes enough monitoring-oriented endpoints to support the first layer of the workflow.	API v2 publicly documents token search, price, ranking, token detail, k-lines, holders, pair transactions, trending, and contract risk reports.	AVE API v2 docs, checked April 10, 2026.	AVE Capability Mapping	Strong
E08A	AVE Sentinel is positioned as a complementary decision layer rather than a replacement for AVE's native browsing surfaces.	Product Docs and the paper introduction state that AVE remains the raw browsing layer for k-lines, holders, transactions, pairs, and risk reports, while Sentinel focuses on SENTINEL-8 scoring, risk gates, evidence explanation, and execution readiness.	Repository inspection, checked Apr. 15, 2026.	Introduction; AVE Capability Mapping; SENTINEL-8	Strong
E09	AVE treats token safety warnings as an actual product concern.	The safety tutorial explains warning surfaces for unaudited and high-risk tokens.	AVE safety tutorial, checked April 10, 2026.	Market and Security Drivers; AVE Mapping	Medium
E10	AVE materials support signal-based opportunity discovery beyond raw token lists.	The Signal Center article uses smart wallet tracking, volume anomalies, market cap acceleration, and social momentum as discovery inputs.	AVE blog, January 11, 2026.	AVE Mapping; Complete Workflow	Medium
E11	AVE also has product-side trading flow evidence.	The Solana trading documentation for AVE Sniper Bot shows fast buy or sell, anti-MEV mode, and post-trade PnL reporting.	AVE Sniper Bot docs, checked April 10, 2026.	AVE Mapping; Complete Workflow	Medium
E12	Solana remains a relevant chain for an early-stage asset system.	DefiLlama reports roughly \$2.882B in 24-hour DEX volume and \$71.967B in 30-day DEX volume.	DefiLlama Solana page, checked April 10, 2026.	Market Drivers	Strong
E13	BSC remains a relevant chain for rapid rotating on-chain opportunities.	DefiLlama reports roughly \$824.94M in 24-hour DEX volume and \$29.768B in 30-day DEX volume.	DefiLlama BSC page, checked April 10, 2026.	Market Drivers	Strong
E14	Meme assets remain large enough to justify a dedicated decision product.	CoinGecko reports roughly \$34.3B in market cap and \$3.19B in 24-hour volume for the meme category.	CoinGecko meme category page, checked April 10, 2026.	Market Drivers	Strong
E15	The BNB meme launch ecosystem has enough scale to matter on BSC.	CoinGecko reports roughly \$748M in market cap for the Four.meme ecosystem.	CoinGecko Four.meme category page, checked April 10, 2026.	Market Drivers	Strong
E16	OKX DEX is a general discovery-plus-execution aggregator.	The official guide describes it as a one-stop, multi-chain and cross-chain trading aggregator with Explore, token profile, liquidity metrics, and broad routing coverage.	OKX DEX user guide, updated March 3, 2026.	Comparative Discussion	Strong
E18	Binance Alpha is positioned as an early-project exposure plus fast-buy entry point.	The official Academy page states that Binance Alpha sits inside Binance Wallet and that Quick Buy includes automated token selection, slippage adjustment, and anti-MEV protection.	Binance Academy, published January 31, 2025 and updated September 11, 2025.	Comparative Discussion	Strong

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ID	Claim	Evidence	Source and Date	Used In	Status
E19	Bitget Onchain mainly reduces the barrier for CEX users to access on-chain assets.	The official support article says users can trade on-chain assets directly in the app using spot-account USDT or USDC across Solana, BSC, and Base.	Bitget support article, April 7, 2025.	Comparative Discussion	Strong
E20	Bybit DEX Pro is best read as a data-enhanced DEX terminal.	Official material highlights influential trader tracking, social checks, real-time trending indicators, wallet integration, cross-chain access, and token safety assessments.	Bybit Learn, May 21, 2024.	Comparative Discussion	Strong
E21	GMGN is strong in memecoin discovery and wallet following, but still leaves safety responsibility to the user.	GMGN docs describe MEME discovery, signals, smart money tracking, and auto-trading, while also warning users to check contract safety before copying trades.	GMGN Quick Start and Track Smart Money docs, checked April 10, 2026.	Comparative Discussion	Strong
E22	GMGN Wallet Radar demonstrates multi-token wallet scanning and copy-trade support.	The official tutorial describes shared-holder discovery, profitable active wallet screening, tracking, and copy trade from the wallet radar surface.	GMGN Wallet Radar docs, checked April 10, 2026.	Comparative Discussion	Strong
E23	Axiom is closer to a Solana execution terminal than to a dual-chain research workflow.	The official FAQ describes Axiom as a hybrid trading app and wallet and states that it currently supports Solana, with more chains planned.	Axiom FAQ, checked April 10, 2026.	Comparative Discussion	Strong
E24	Axiom already makes structural token filtering first-class.	The Explore Tokens docs show filters for top-holder percentage, liquidity, volume, and transaction count on trending and new-pairs pages.	Axiom Explore Tokens docs, checked April 10, 2026.	Comparative Discussion	Strong
E25	Axiom also supports wallet-level trade interpretation for memecoins.	Trader Scan shows buys, sells, current balance, realized PnL, and hold time for wallets interacting with a token.	Axiom Trader Scan docs, checked April 10, 2026.	Comparative Discussion	Strong
E26	Security conditions are severe enough to justify central risk gating.	CertiK estimates about \$3.352B in 2025 losses and about \$1.451B in supply chain attack losses.	CertiK 2025 report, December 23, 2025.	Market and Security Drivers	Strong
E27	Scam pressure is still worsening at the broader market level.	Chainalysis estimates about \$17B in scam and fraud losses for 2025 and about 1400% growth in impersonation scams.	Chainalysis, January 13, 2026.	Market and Security Drivers	Strong
E28	The current opportunity and risk case studies are prototype walkthroughs rather than live-performance claims.	The mock dataset encodes one positive case and one blocked case, while replay data can also receive event records written from Opportunity Desk actions.	Repository inspection, checked Apr. 15, 2026.	Case Studies; Replay	Medium
E29	The paper's Solana+BSC narrative is narrower than the current runtime surface.	Product copy still defines Solana+BSC as the main story, but current code already exposes Radar and trade tooling for Base and Ethereum as well.	Repository inspection, checked Apr. 15, 2026.	Radar; Limitations	Strong

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ID	Claim	Evidence	Source and Date	Used In	Status
E30	The claim that AVE Sentinel is "more complete" is grounded in workflow closure rather than absolute performance claims.	The interface and repository documents connect Overview, Docs, Access, Radar, Score Model, Dossier, Risk, Opportunity, embedded replay, CLI, Telegram, and the skill runtime into one task chain, while competing official materials concentrate on subsets of that chain.	Repository inspection plus official competitor pages, synthesis checked Apr. 15, 2026.	Introduction; Conclusion	Medium
E31	CLI and Telegram are not separate prototypes but synchronized entry points.	The shared `sentinel_core` implementation serves Radar, brief, token, risk, quote, wallet, approval, and order outputs to both CLI and Telegram, reusing the same upstream logic.	Repository inspection, checked Apr. 15, 2026.	Architecture; Complete Workflow	Strong
E32	SENTINEL-8 is presented as an eight-signal latent risk-gated scoring model rather than a rhetorical label.	The paper and repository together support an interpretable eight-signal layer, a compact latent evidence-view formulation, risk-aware score suppression, and explicit `Proceed / Watch / Block` decision semantics.	Repository inspection, checked Apr. 15, 2026.	SENTINEL-8 section	Strong
E33	The skill layer is a first-class structured runtime rather than an internal helper.	The workspace provides skill metadata, explicit JSON schemas, grouped operations for `data`, `chain_wallet`, and `delegate_wallet`, and a stable response envelope with shared fields.	Repository inspection, checked Apr. 15, 2026.	Architecture; Skill Runtime; Complete Workflow	Strong
E34	The frontend already exposes a multi-chain wallet-connect surface through Privy-based onboarding.	The UI mounts a wallet-connect button that lazy-loads a Privy provider, configures wallet-only login, supports both EVM and Solana connectors, and offers switch or disconnect handling after connection.	Repository inspection, checked Apr. 15, 2026.	Opportunity Desk; Limitations	Strong
E35	The same decision model is now reusable across human-facing and agent-facing surfaces.	Web, CLI, Telegram, and the skill runtime all read from the same shared core or the same scoring and trade layers, so latent score, confidence, verdict, and execution-readiness semantics can be described as one surface-invariant contract.	Repository inspection, checked Apr. 15, 2026.	SENTINEL-8; Complete Workflow; Conclusion	Medium

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